INTELLIGENT AGENTS AND KNOWLEDGE RESPRESENTATION IN ARTIFICIAL INTELLIGENCE

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Outline

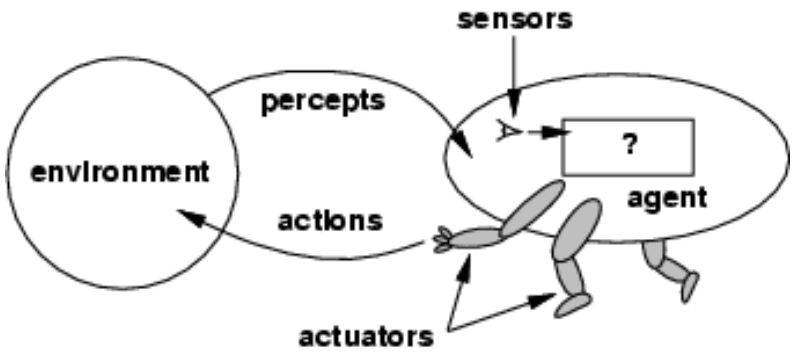
- Agents
- Knowledge Representation and Reasoning
  - ✓ Perception
  - ✓ Learning
  - ✓ Reasoning
  - ✓ Planning
  - ✓ Execution
- Propositional logic
- First-order logic
- Other Logics

Intelligent Agents

Definition: An **intelligent agent** is a system that perceives its environment, processes information, and takes actions autonomously to achieve specific goals or objectives.

- An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators
- Human agent: eyes, ears, and other organs for sensors; hands, legs, mouth, and other body parts for actuators
- Robotic agent: cameras and infrared range finders for sensors; various motors for actuators

Agents and environments



- The agent function maps from percept histories to actions:

$$[f: \mathcal{P}^{\star} \rightarrow \mathcal{A}]$$

- The agent program runs on the physical architecture to produce f

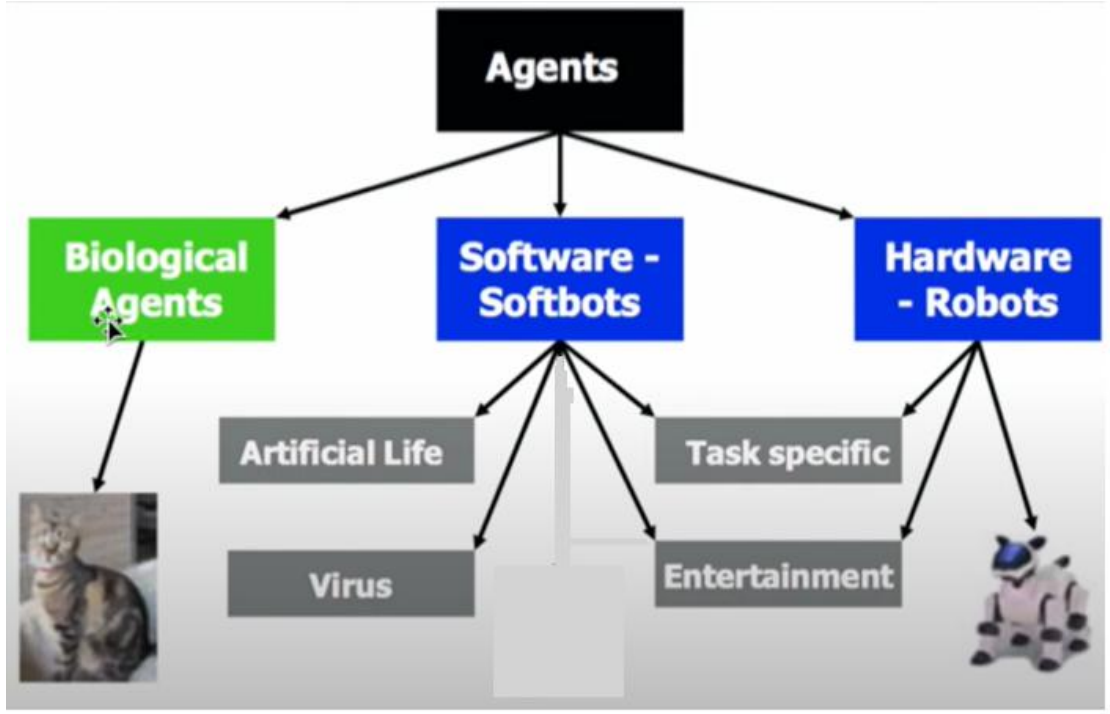
- agent = architecture + program

Agents Application

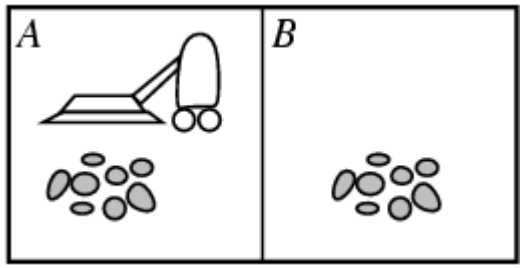
- Human Agents: Biological entities, using cognitive abilities such as reasoning, memory, and decision-making to interact with the environment.

- Software Agents: Digital entities operating in virtual environments, performing tasks autonomously. Software agents can analyze data, automate tasks, interact with other software systems, and make decisions based on predefined rules or learning algorithms.

- Robotic Agents: Physical entities operating in the real world, performing tasks through physical actions. They combine sensors, actuators, and AI to perceive the environment and perform physical actions. These agents can range from simple machines to highly complex autonomous robots.



Vacuum-cleaner world



- Percepts: location and contents, e.g., [A, Dirty]
- Actions: *Left*, *Right*, *Clean*, *NoOp*

Rational agents

- An agent should strive to "do the right thing", based on what it can perceive and the actions it can perform. The right action is the one that will cause the agent to be most successful

- Performance measure: An objective criterion for success of an agent's behavior

- E.g., performance measure of a vacuum-cleaner agent could be amount of dirt cleaned up, amount of time taken, amount of electricity consumed, amount of noise generated, etc.

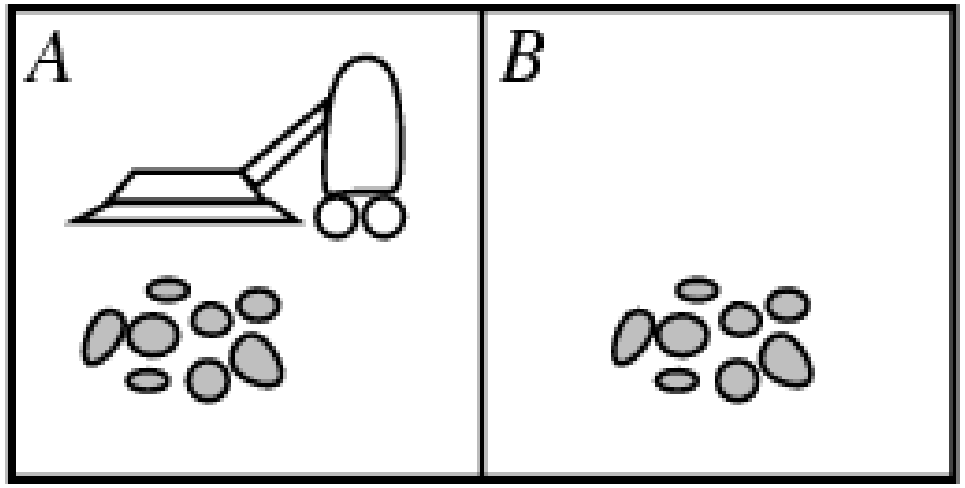
Rational agents

- Rational Agent: For each possible percept sequence, a rational agent should **select an action** that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.

Rational agents

- Agents can perform actions in order to modify future percepts so as to obtain useful information (information gathering, exploration)

- An agent is autonomous if its behavior is determined by its own experience (with ability to learn and adapt)



Percept sequence	Action
$[A, Clean]$	$Right$
$[A, Dirty]$	$Clean$
$[B, Clean]$	$Left$
$[B, Dirty]$	$Clean$

What are the key aspects of Intelligent agent?

1. Autonomy

- **Definition**: The agent's ability to operate **independently** without human intervention.
- **Importance**: Autonomy allows the agent to make decisions and take actions on its own, based on the information it gathers from its environment.

2. Perception

- **Definition**: The ability of the agent to **sense and interpret data** from its environment.
- **Importance**: Perception is critical for understanding the environment in which the agent operates. It involves using sensors or data inputs to gather information and interpreting this data to form a coherent understanding of the current state.

3. Learning and Adaptation

- **Definition**: The agent's capacity to **learn from experience and improve its performance** over time.
- **Importance**: Learning allows the agent to adapt to new situations or changes in its environment, improving its effectiveness and efficiency in achieving its goals.

4. Reasoning and Decision-Making

- **Definition**: The process by which the agent **evaluates information, considers possible actions, and chooses the best course of action**.
- **Importance**: Intelligent agents need to make decisions that align with their objectives, taking into account current conditions, possible outcomes, and constraints.

5. Actuation

- **Definition**: The mechanisms or actions the agent can use **to interact with** and influence its environment.
- **Importance**: Actuators are what allow the agent to implement its decisions and make changes in the environment, whether through physical movements, sending information, or other actions.

6. Communication

- **Definition**: The ability of the agent to **communicate** with other agents.
- **Importance**: Effective communication allows the agent to share information, coordinate actions, and collaborate with other entities. This is especially important in multi-agent systems or human-agent interactions.

7. Goal-Oriented Behavior

- **Definition**: The focus of the agent on **achieving** specific objectives or goals.
- **Importance**: Goal-oriented behavior ensures that the agent's actions are purposeful and directed towards fulfilling its mission, whether that involves maximizing efficiency, solving a problem, or achieving a set of defined outcomes.

8. Robustness

- **Definition**: The ability of the agent to **handle** unexpected situations and remain effective under varying conditions.
- **Importance**: Flexibility allows the agent to adapt to changes or new information, while robustness ensures that the agent can continue functioning even when faced with challenges or errors.

9. Memory

- **Definition**: The agent's ability to **store** and **recall past** information.
- **Importance**: Memory allows the agent to use historical data for learning, decision-making, and predicting future events.

What factors should we consider when designing an intelligent agent?

PEAS

- PEAS: Performance measure, Environment, Actuators, Sensors
- Must first specify the setting for intelligent agent design
- Consider, e.g., the task of designing an automated taxi driver:
  - Performance measure
  - Environment
  - Actuators
  - Sensors

PEAS

- Must first specify the setting for intelligent agent design
- Consider, e.g., the task of designing an automated taxi driver:
  - Performance measure: Safe, fast, legal, comfortable trip, maximize profits
  - Environment: Roads, other traffic, pedestrians, customers
  - Actuators: Steering wheel, accelerator, brake, signal, horn
  - Sensors: Cameras, sonar, speedometer, GPS, odometer, engine sensors, keyboard

PEAS

- Agent: Medical diagnosis system
- Performance measure: Healthy patient, minimize costs, lawsuits
- Environment: Patient, hospital, staff
- Actuators: Screen display (questions, tests, diagnoses, treatments etc.)
- Sensors: Keyboard (entry of symptoms, findings, patient's answers)

PEAS

- Agent: Part-picking robot
- Performance measure: Percentage of parts in correct bins
- Environment: Conveyor belt with parts, bins
- Actuators: Jointed arm and hand
- Sensors: Camera, joint angle sensors



Environment types

Fully observable (vs. partially observable): An agent's sensors give it access to the complete state of the environment at each point in time.

Deterministic (vs. stochastic): Deterministic are the environments where the next state is observable at a given time. So there is no uncertainty in the environment. The Stochastic environment is the opposite of a deterministic environment. The next state is totally **unpredictable** for the agent.

Episodic (vs. sequential): The agent's experience is divided into atomic "episodes" (each episode consists of the agent perceiving and then performing a single action), and the choice of action in each episode depends only on the episode itself.

Environment types

Static (vs. dynamic): The environment is unchanged while an agent is deliberating. (The environment is semidynamic if the environment itself does not change with the passage of time but the agent's performance score does)

Discrete (vs. continuous): A limited number of distinct, clearly defined percepts and actions.

Single agent (vs. multiagent): An agent operating by itself in an environment.

Environment types

	Chess without a clock	Taxi driving
Fully observable	Yes	No
Deterministic	Yes	No
Episodic	Yes	No
Static	Yes	No
Discrete	Yes	No
Single agent	No	No

- The environment type largely determines the agent design
- The real world is (of course) partially observable, stochastic, sequential, dynamic, continuous, multi-agent

https://www.youtube.com/watch?v=_Y5_iGxWFrQ&t=81s

Cont.

- not fully observable: Limited Sensor Range and Field of View
- not deterministic: **Unpredictable Opponent Behavior**: The actions of opposing robots are not predictable.
- a RoboCup field is static itself but since the robots dynamically move on the field it is **not fully static**. The robots themselves are dynamic as they move around the field, changing their positions, orientations, and velocities based on their actions and strategies.
- In RoboCup, the field where autonomous robots play soccer is not fully discrete, but rather a blend of discrete and continuous elements. **Game Time**: The game is divided into discrete periods, such as halves or quarters, which are finite and clearly defined. **Movement and Positioning**: The robots and the ball move continuously across the field.
- Multiagents

Design and create an intelligent agent to clean the dirty house.

Example



Example: Demonstration of an intelligent agent

- **Environment**: A living room is divided into 5x5 parts and 10 parts of the living room initially containing dirt (randomly generated).
- **Agent's Task**: The agent moved around the living room, detecting and cleaning dirt until all parts of the living room are clean.
- **Movement**: The agent moves one step at a time (up, down, left, or right) toward the nearest dirt.
- **Cleaning**: When the agent reaches a part with dirt, it immediately cleans the part in the living room. The dirt is removed (D turns into ‘.’), and the living room is displayed again to reflect the change. When the agent not detect dirt, and no action is required.
- **Visualization**: The agent's position is shown with A, dirt is shown with D, and clean parts are shown with ‘.’
- **Result**: The grid is now completely clean, and the agent cleaned a total of 10 parts of the living room.

Agent functions and programs

An agent is completely specified by the agent function mapping percept sequences to actions

One agent function (or a small equivalence class) is rational

Aim: find a way to implement the rational agent function concisely

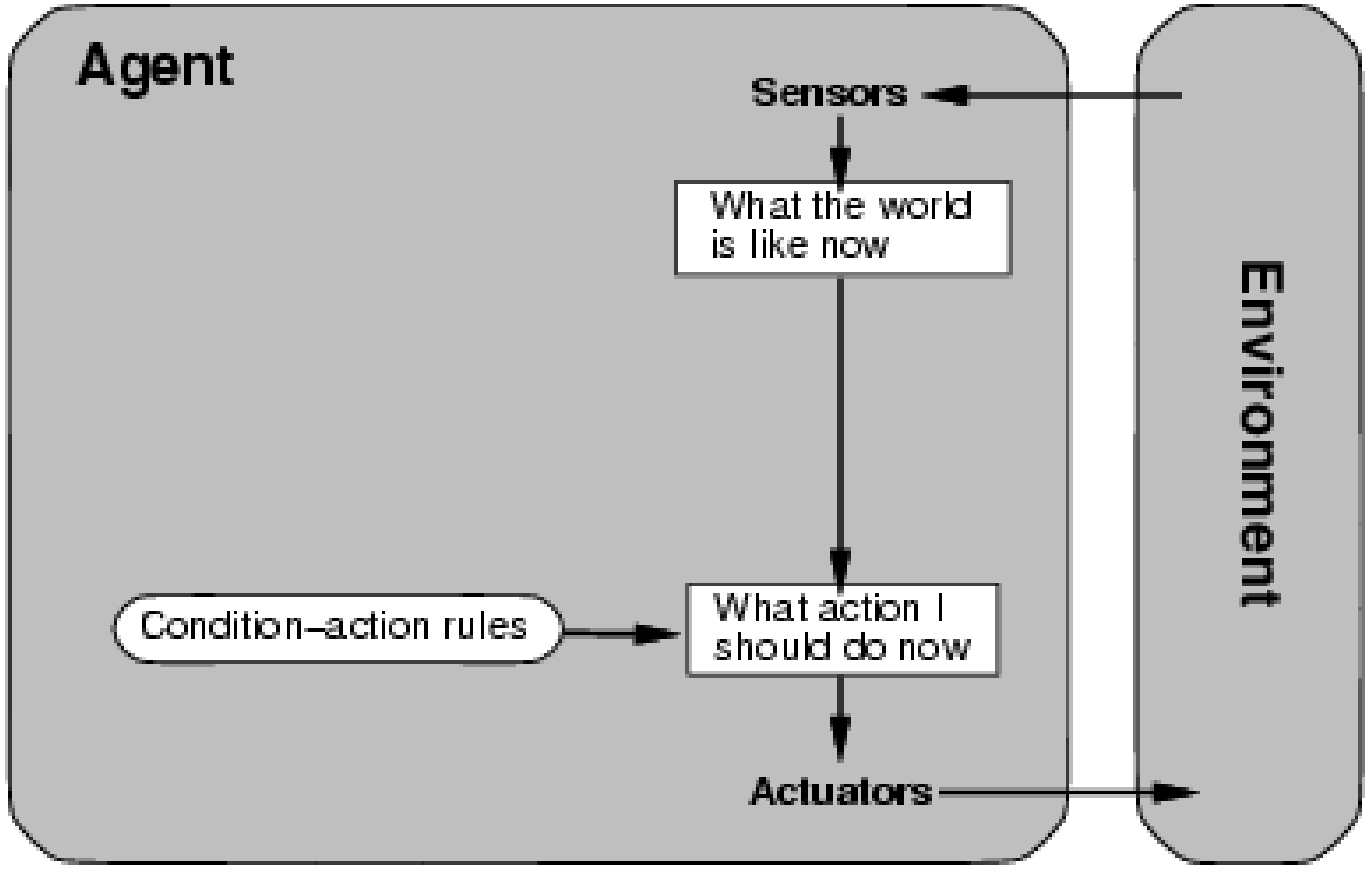
Agent types

- Four basic types in order of increasing generality:

  - ✓ Simple reflex agents
  - ✓ Model-based reflex agents
  - ✓ Goal-based agents
  - ✓ Utility-based agents

Simple reflex agents

Simple Reflex Agents are a fundamental type of agent in artificial intelligence (AI) that operate based on a straightforward principle.



Percept: The agent perceives information from its environment. This could be anything relevant to its task, like sensors detecting obstacles or changes in the environment.

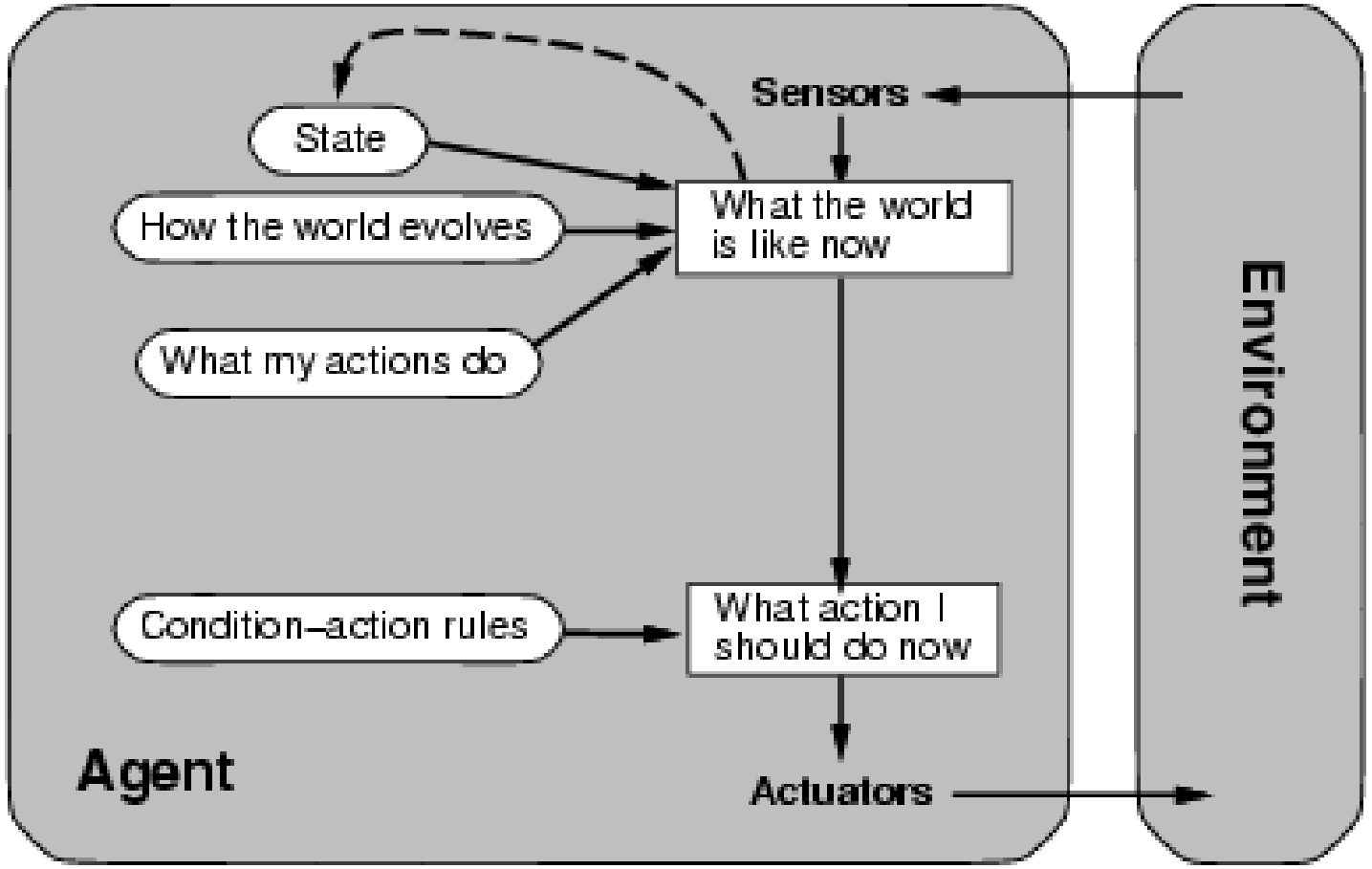
Condition-Action Rules: Simple Reflex Agents use a set of **condition-action rules**, also known as production rules or "**if-then" rules**. These rules specify the actions to take in response to particular conditions or percepts. For example, a rule might state, "If an obstacle is detected in front, then turn left."

Action: Based on the current percept and the matching rule, the agent performs a specific action. The agent does not consider the history of past actions or percepts; it only reacts to the current situation.

No Memory or Learning: Simple Reflex Agents do not have memory or learning capabilities. They do not use past experiences or adapt their behavior based on previous interactions.

Model-based reflex agents

Model-based reflex agents are a more advanced type of agent compared to simple reflex agents.



Internal Model: Unlike simple reflex agents, which only react to the current percept, model-based reflex agents maintain an internal model of the world. This model represents the agent's understanding of the environment and its own state. The model helps the agent keep track of aspects of the environment that are not immediately visible or perceptible.

Percept: The agent still receives percepts or inputs from the environment, which provide information about the current state of the environment.

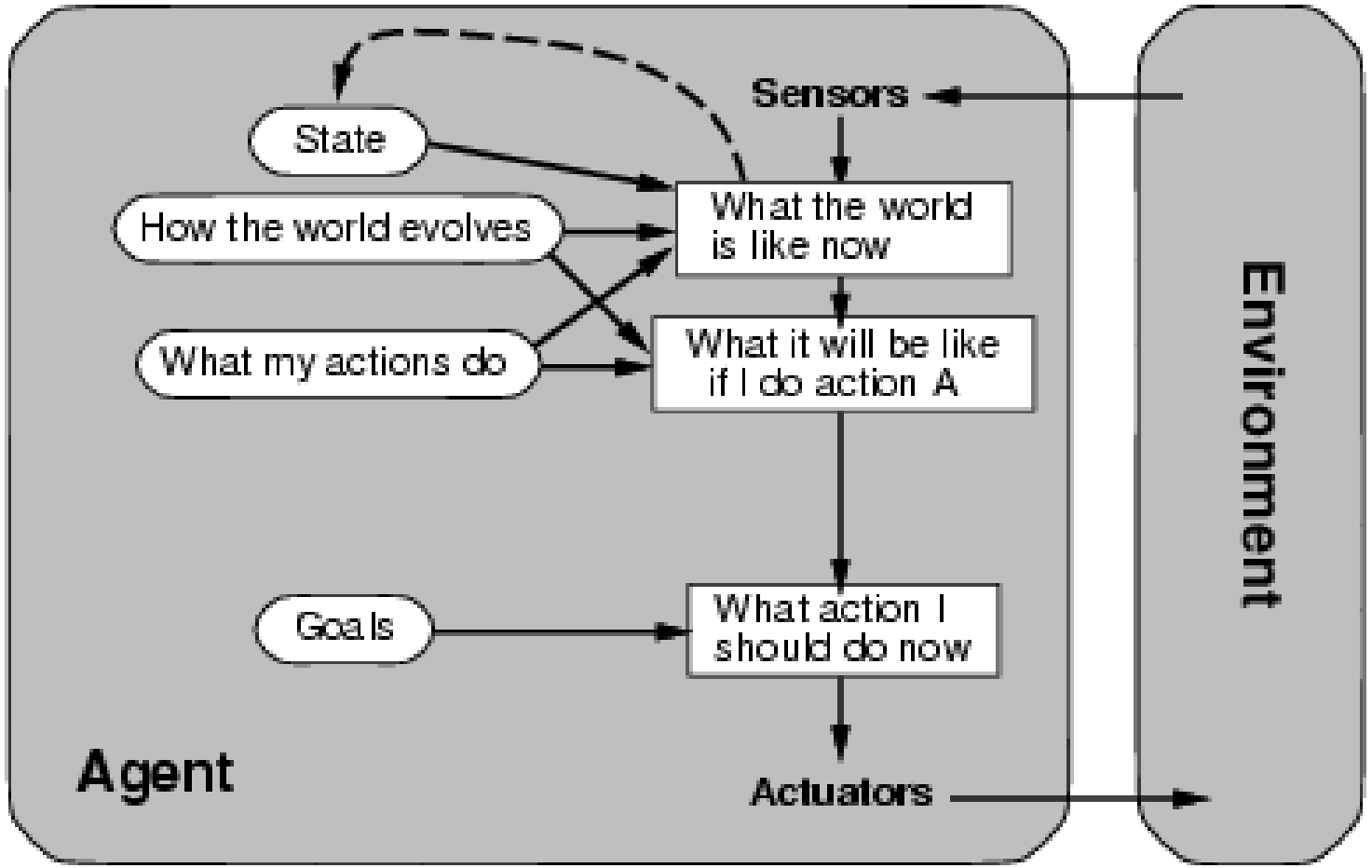
Model of the World: This model keeps track of the state of the environment and the agent's state over time. It includes information about how the environment changes and how the agent's actions affect the environment.

Condition-Action Rules: The agent uses these rules to decide on actions based on its current percept and the internal model. The rules might be more complex than those of a simple reflex agent because they can take the model's state into account.

Action: Based on the percept, the internal model, and the condition-action rules, the agent performs an action.

Goal-based agents

Goal-based agents represent a more advanced class of agents in artificial intelligence. Unlike simple reflex agents or model-based reflex agents, which react to current percepts or maintain an internal model of the world, goal-based agents are designed to act in a way that achieves specific goals.



Goals: Goal-based agents operate with specific objectives or goals they aim to achieve.

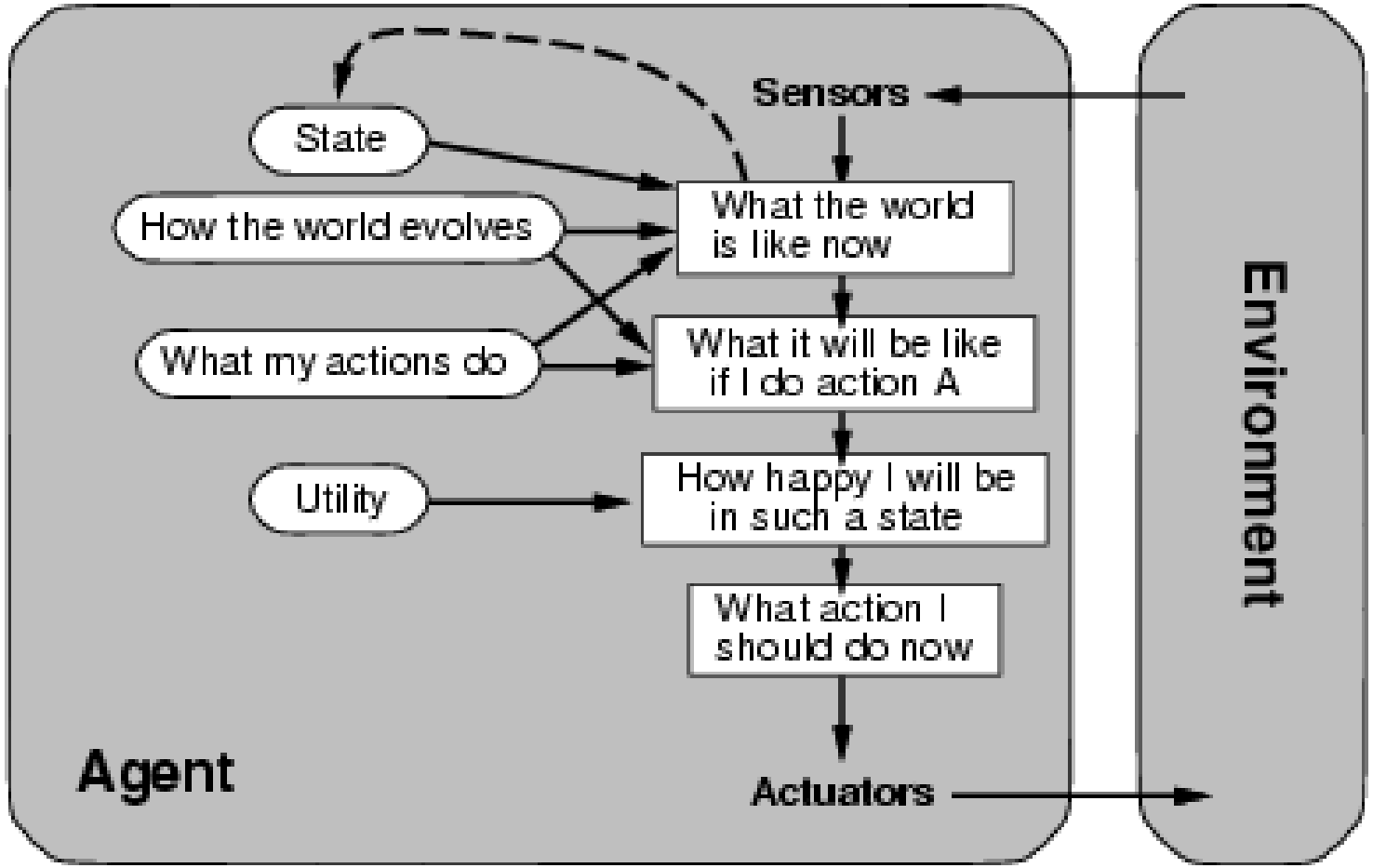
Planning: To achieve their goals, these agents typically involve a planning process. They create a sequence of actions or a strategy that will lead them from their current state to a desired goal state. This often requires evaluating different possible actions and selecting the most appropriate ones to reach the goal.

State Space: Goal-based agents operate within a state space, which is a representation of all possible states the environment and the agent can be in. They use this state space to evaluate how their actions will affect their progress towards achieving their goals.

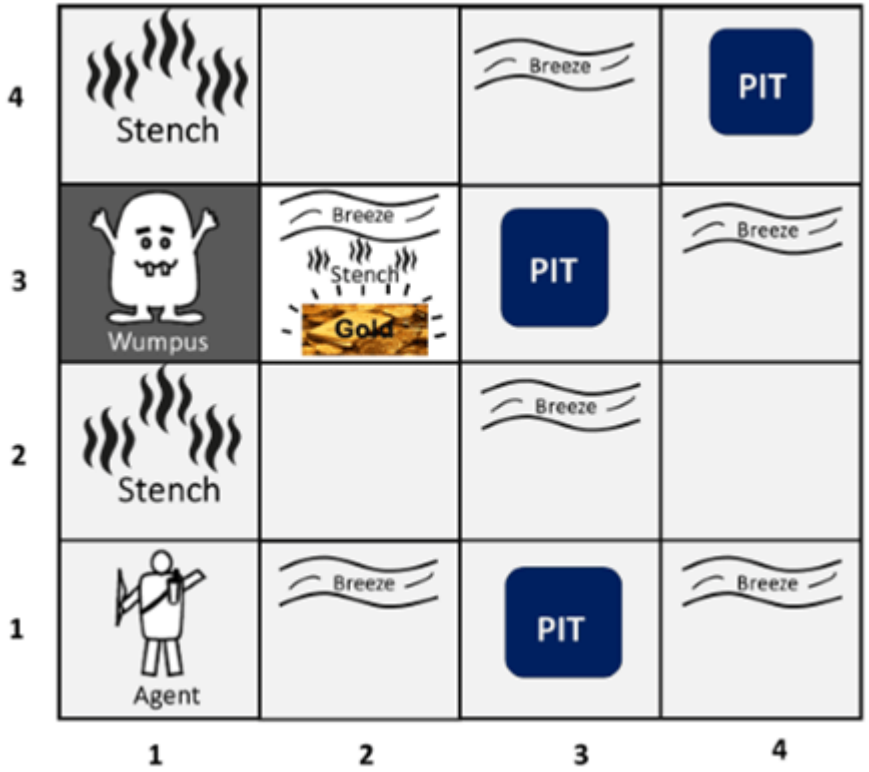
Decision Making: The agent decides on actions based on their potential to move the agent closer to the goal. This decision-making process involves considering the current state, the possible actions, and how those actions will influence the state in relation to the goal.

Utility-based agents

not just aiming to achieve specific goals but also maximizing a measure of "utility" or "satisfaction" derived from their actions



Example: Wumpus World



1.The rooms adjacent to the Wumpus room are smelly, so that it would have some stench.

2.The room adjacent to PITs has a breeze, so if the agent reaches near to PIT, then he will perceive the breeze.

3.There will be glitter in the room if and only if the room has gold.

4.The Wumpus can be killed by the agent if the agent is facing to it, and Wumpus will emit a horrible scream which can be heard anywhere in the cave.

Wumpus World

Environment:

- A 4*4 grid of rooms.
- The agent initially in room square [1, 1], facing toward the right.
- Location of Wumpus and gold are chosen randomly except the first square [1,1].
- Each square of the cave can be a pit with probability 0.2 except the first square.

Actuators:

- Left turn,
- Right turn
- Move forward
- Grab

Properties of the Environment

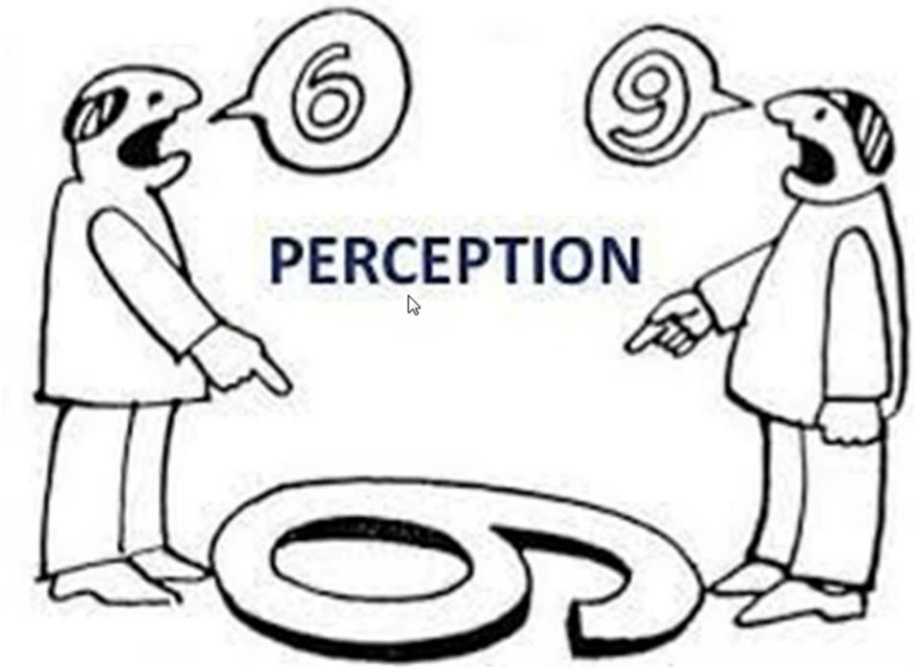
- **Partially observable:** The Wumpus world is partially observable because the agent can only perceive the close environment such as an adjacent room.
- **Deterministic:** It is deterministic, as the result and outcome of the world are already known.
- **Static:** It is static as Wumpus and Pits are not moving.
- **Discrete:** The environment is discrete.
- **One agent:** The environment is a single agent as we have one agent only and Wumpus is **not** considered as an agent.

Knowledge Representation and Reasoning

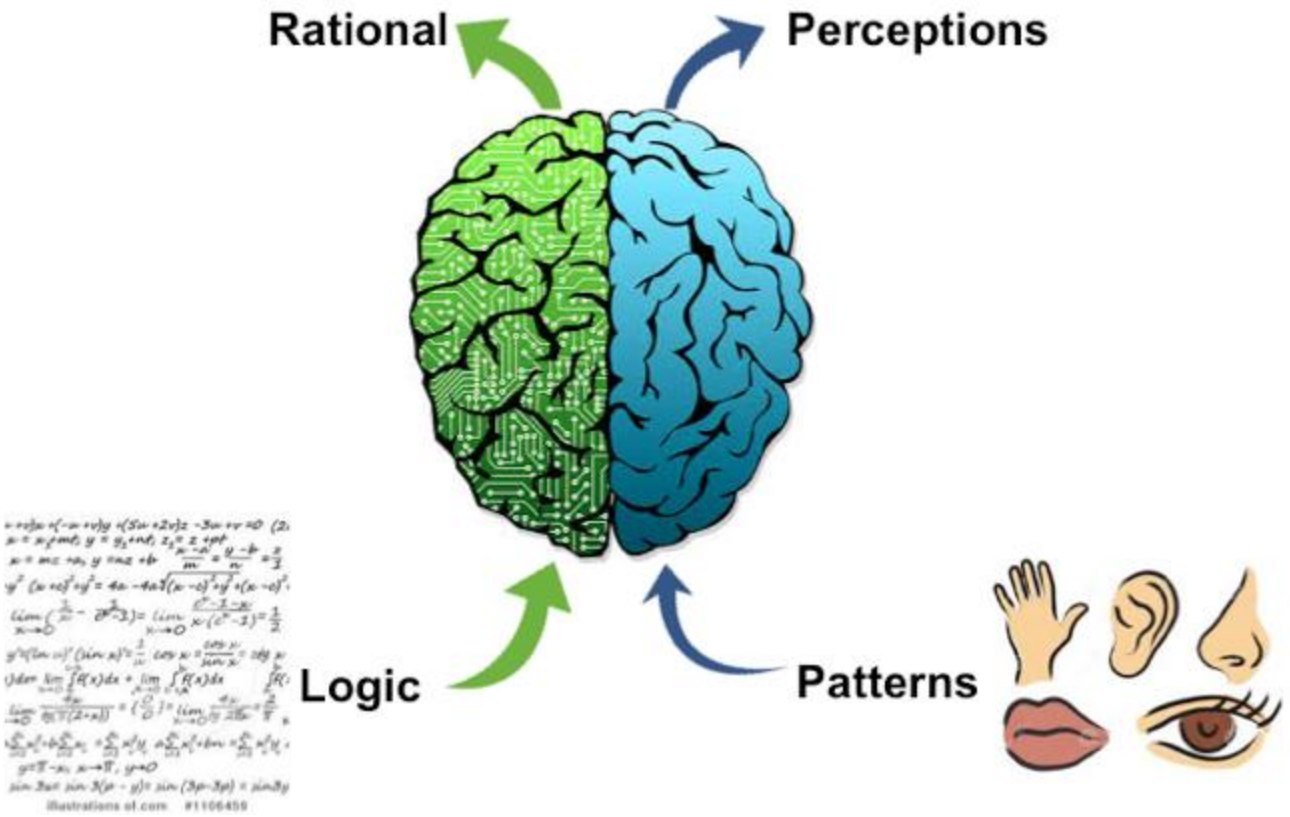
What is knowledge?

- Knowledge plays a vital role in intelligence as well as creating artificial intelligence.

- It demonstrates the intelligent behavior in AI agents or systems.



Human-Intelligence (Knowledge Representation)



Cont.

- Knowledge representation (KR) is the study of
  - how knowledge and facts about the world can be represented, and
  - what kinds of reasoning can be done with that knowledge.



Why is it important?

- Knowledge Representation (KR) is about how we formally store and structure information so that an AI system can reason, learn, and make decisions effectively.

1. Bridges Human Knowledge and Machines
Humans understand the world in terms of facts, concepts, and relationships (e.g., "Cats are animals" or "Water freezes below 0°C").
Machines need this information in a structured format so they can process it (like graphs, logic, or rules).

2. Enables Reasoning and Inference
AI can *infer new facts* from existing knowledge if it's represented logically.
Example: If we store "All humans are mortal" and "Socrates is a human," AI can infer "Socrates is mortal."

Cont.

3. Supports Problem Solving

•Representation helps AI model real-world situations (e.g., in planning, medical diagnosis, or game strategies).

•Instead of memorizing data, the AI can **reason about actions, consequences, and alternatives.**

4. Makes Learning Possible

•Knowledge structures guide machine learning systems by providing **background context**.

•Example: A self-driving car learns faster if it already has knowledge of road rules and object categories.

Knowledge representation: goals

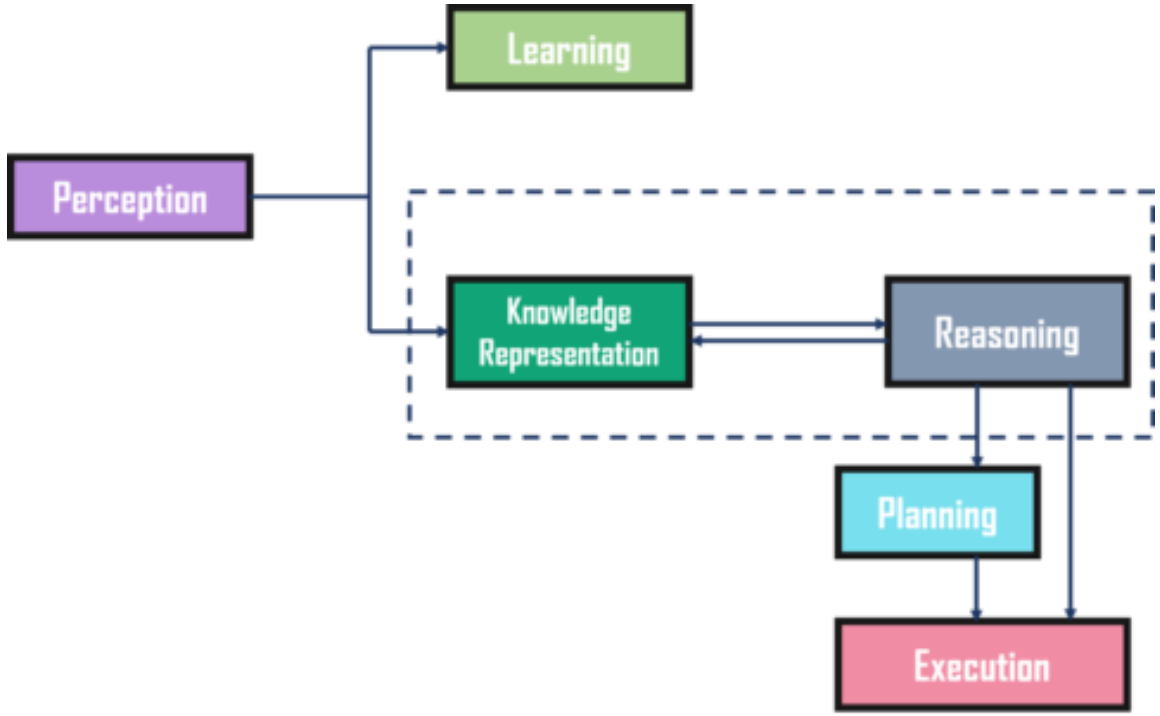
We want a representation that is:

- rich enough to express the knowledge needed to solve the problem
- compact, natural and maintainable, amenable to efficient computation
- able to express features
- able to be used for trade off accuracy and computation time

Cycle of Knowledge Representation

- The **Perception component** retrieves data or information from the environment. With the help of this component, you can retrieve data from the environment.

- The **Learning Component** that learns from the captured data by the perception component.
- The goal is to build computers that can be taught.
- In order to learn new things, the system requires knowledge acquisition, inference, acquisition of heuristics, faster searches, etc.



How everything works together
A simple way to explain the whole diagram to students is:
Perception → Understand the environment
Learning → Improve from experience
Knowledge Representation → Store what is known
Reasoning → Decide what follows from what is known
Planning → Decide what actions to take
Execution → Perform those actions

Cont.

- The main component in the cycle is **Knowledge Representation and Reasoning** which shows the human-like intelligence in the machines.

- Knowledge representation is all about understanding intelligence

- its goal is to understand and build intelligent behavior from the top-down and focus on what an agent needs to know in order to behave intelligently.
- The **Planning and Execution** components depend on the analysis of knowledge representation and reasoning.
  - planning includes giving an initial state, finding their preconditions and effects, and a sequence of actions to achieve a state in which a particular goal holds.
  - the final stage is the execution of the entire process.

Approaches to Knowledge Representation

Logical Representation

Logical Representation

- The simplest kind of logical representation is propositional logic (PL), in which all statements are made up of propositions.

- In general, propositions are statements that are either **TRUE** or **FALSE**.

Cont.

- It operates with 0 and 1, propositional logic is also known as Boolean logic.
- In propositional logic, symbolic variables are used to express the logic, and any symbol can be used to represent a proposition, such as A, B, C, P, Q, R, and so on.
- Propositions can be true or untrue, but **not** both at the same time.
- An object, relations or functions, and **logical connectives** make up propositional logic.
- Logical operators are another name for these connectives.

Examples

- Following are examples of propositions that evaluate to TRUE

There are seven days in a week

Stockholm is the capital city of Sweden

4+2= 6

Examples

- Following propositions evaluate to FALSE

Sun is cold

London is the capital of France

2 – 4 = 7

Logical connectives

- Propositions can be combined into compound statements by operators called **logical connectives**;
- The purpose of defining these connectiveness is to provide precise meanings to such words as “**and**” and “**or**” that occur in the natural language;
- The way we give semantic meaning to these connectives is to provide **tables** known as **truth tables**;
- These give a value for every possible combination of the values of the individual statements that make up the compound proposition.

Symbols

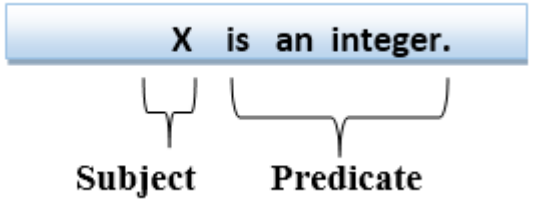
Connective symbols	Word	Technical term	Example
∧	AND	Conjunction	A ∧ B
∨	OR	Disjunction	A ∨ B
→	Implies	Implication	A → B
⇔	If and only if	Biconditional	A⇔ B
¬ or ~	Not	Negation	¬ A or ¬ B

First-Order Logic

is a logical system for reasoning about properties of objects.

First-order logic statements can be divided into two parts:

- **Subject:** is the main part of the statement.
- **Predicate:** can be defined as a relation between words in a statement.
- **Consider the statement: "x is an integer."**, it consists of two parts, the first part x is the **subject** of the statement and second part "is an integer," is known as a **predicate**.



Example:

Ravi and Ajay are brothers: => Brothers(Ravi, Ajay).
Chinky is a cat: => cat (Chinky).

We can represent such sentences as **Predicate (term1, term2, ..., term n)**.

Example

$$Likes(You, Eggs) \land Likes(You, Tomato) \rightarrow Likes(You, Shakshuka)$$



$$Likes(You, Eggs) \wedge Likes(You, Tomato) \rightarrow Likes(You, Shakshuka)$$

These blue terms are called *constant symbols*. Unlike propositional variables, they refer to *objects*, not *propositions*.

The red things that look like function calls are called *predicates*. Predicates take objects as arguments and evaluate to true or false.

Examples:

1. Lucy* is a professor
2. John is the dean.

is-prof(lucy)

is-dean(John)

The Existential Quantifier

- A statement of the form

  $$\exists x.\ \textit{some-formula}$$

  is true if, for *some* choice of x, the statement ***some-formula*** is true when that x is plugged into it.

- Examples:

  $\exists x.\ (\mathit{Even}(x) \wedge \mathit{Prime}(x))$

Example

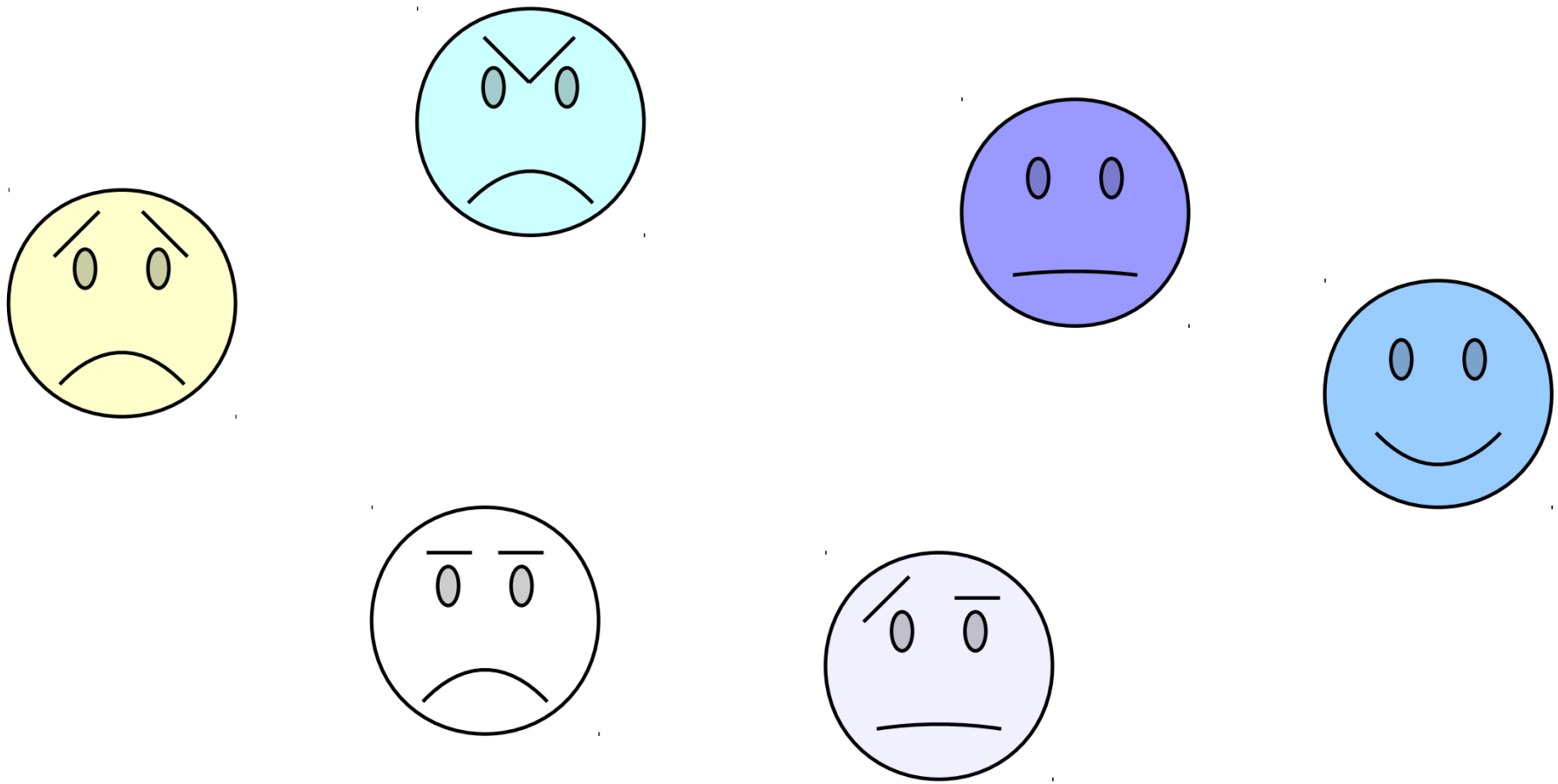
Some boys play cricket.

In this question, the predicate is "**play(x, y)**," where x= boys, and y= game.

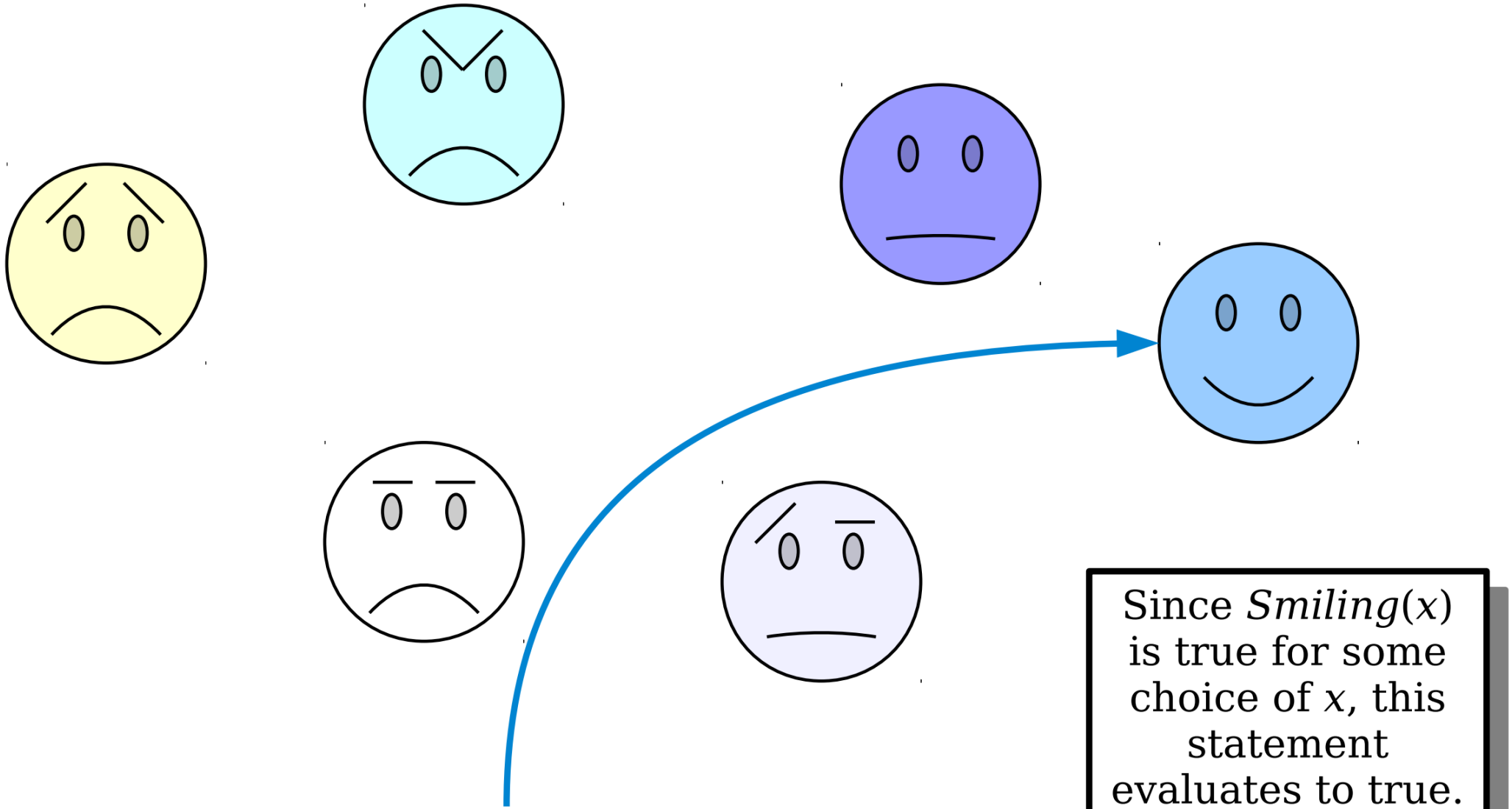
Since there are some boys so we will use **∃, and it will be represented as**:

∃x boys(x) → play(x, cricket).

If there are some x who are boys then they play cricket.

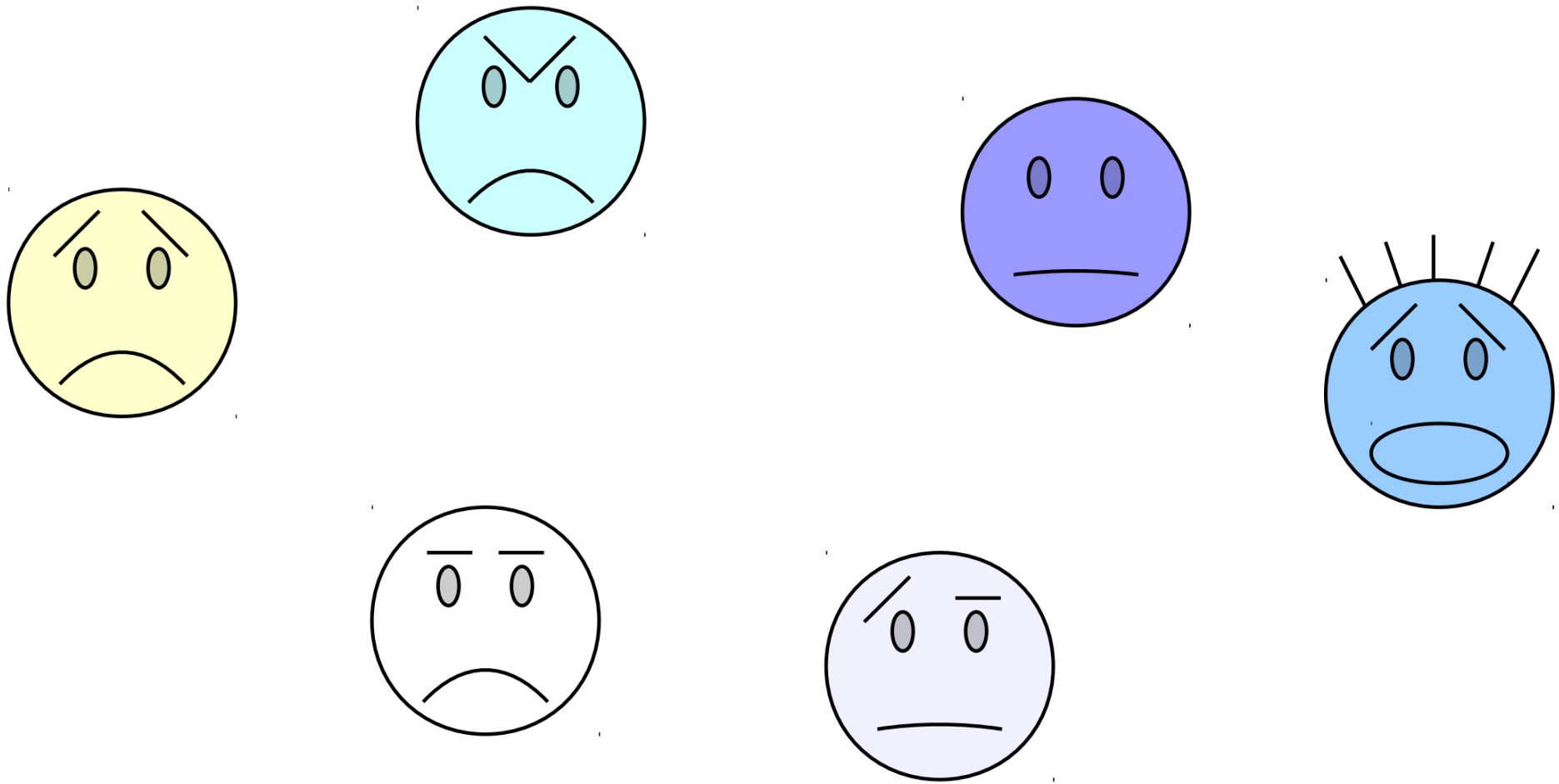


$\exists x.\ \mathit{Smiling}(x)$

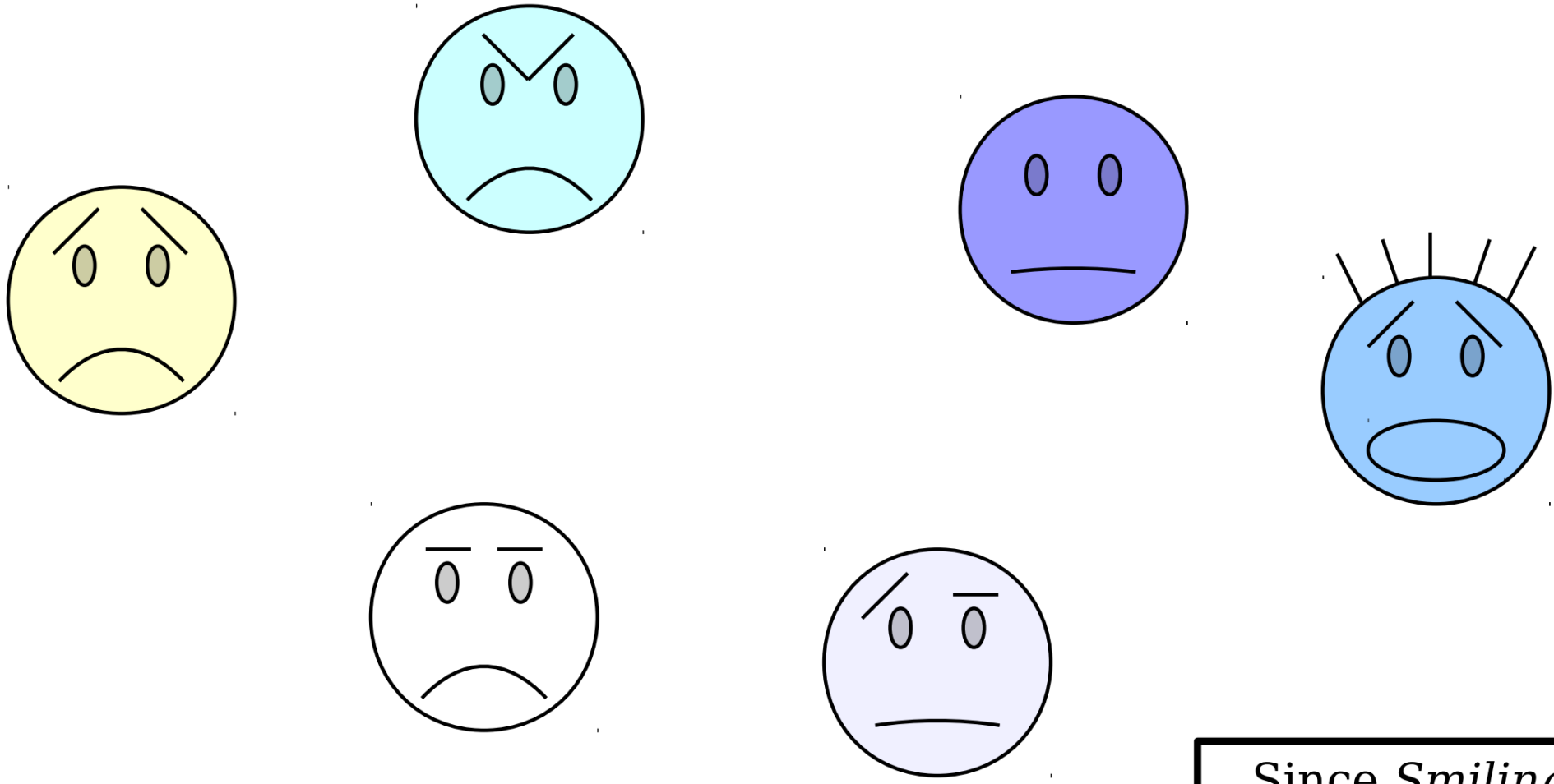


$\exists x.\ Smiling(x)$

Since $Smiling(x)$ is true for some choice of x, this statement evaluates to true.



$\exists x.\ Smiling(x)$



$\exists x.\ Smiling(x)$

Since $Smiling(x)$ is not true for any choice of x, this statement evaluates to false.

The Universal Quantifier

- A statement of the form

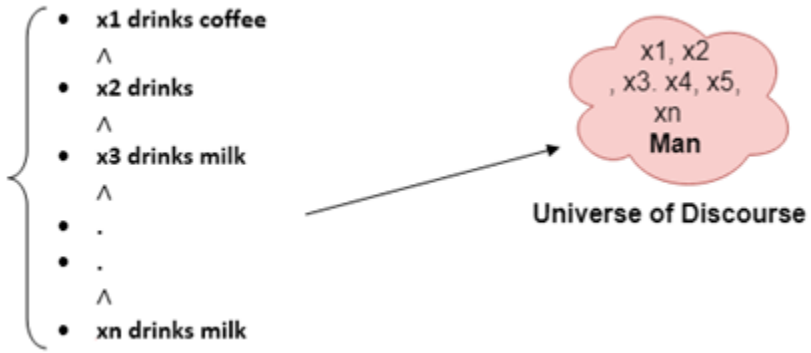
  $\forall x.$

  is true if, for every choice of x, the statement is true when x is plugged into it.

Example:

All man drink coffee**.**

∀x man(x) → drink (x, coffee).
It will be read as: There are all x where x is a man who drink coffee.

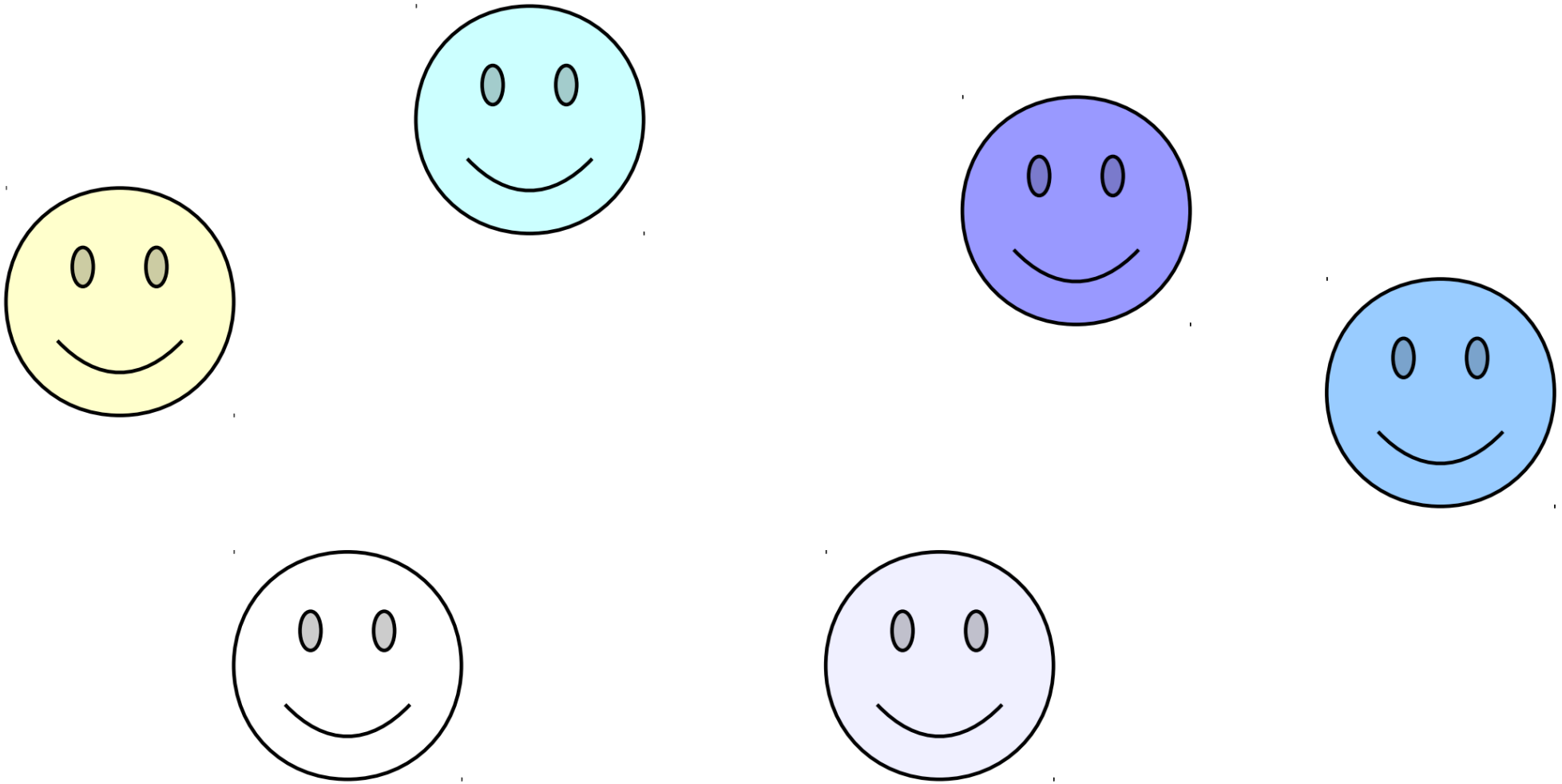


Example

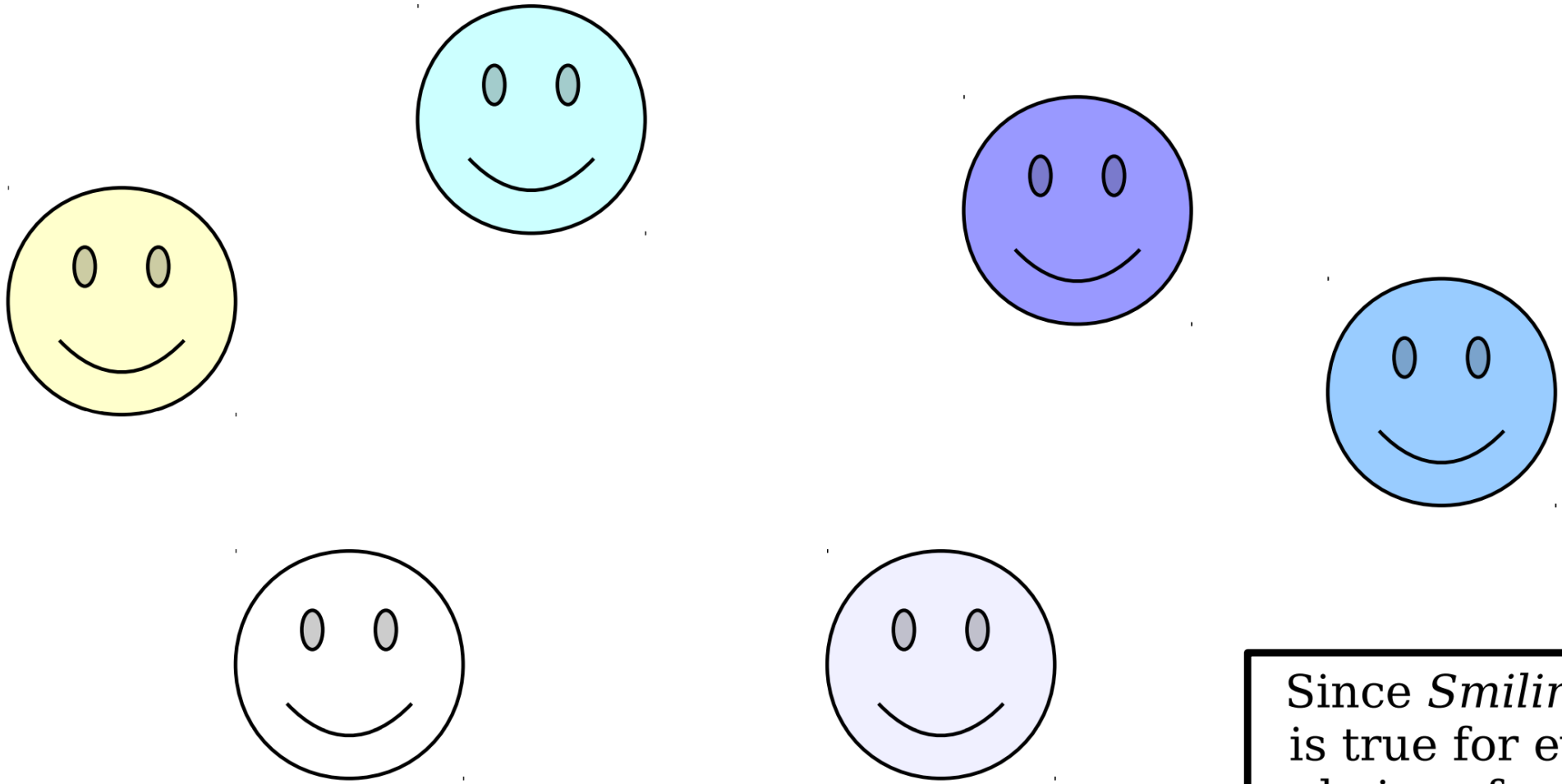
- **1. All birds fly.**

In this question the predicate is "**fly(bird)**."
And since there are all birds who fly so it will be represented as follows.

∀x bird(x) → fly(x).

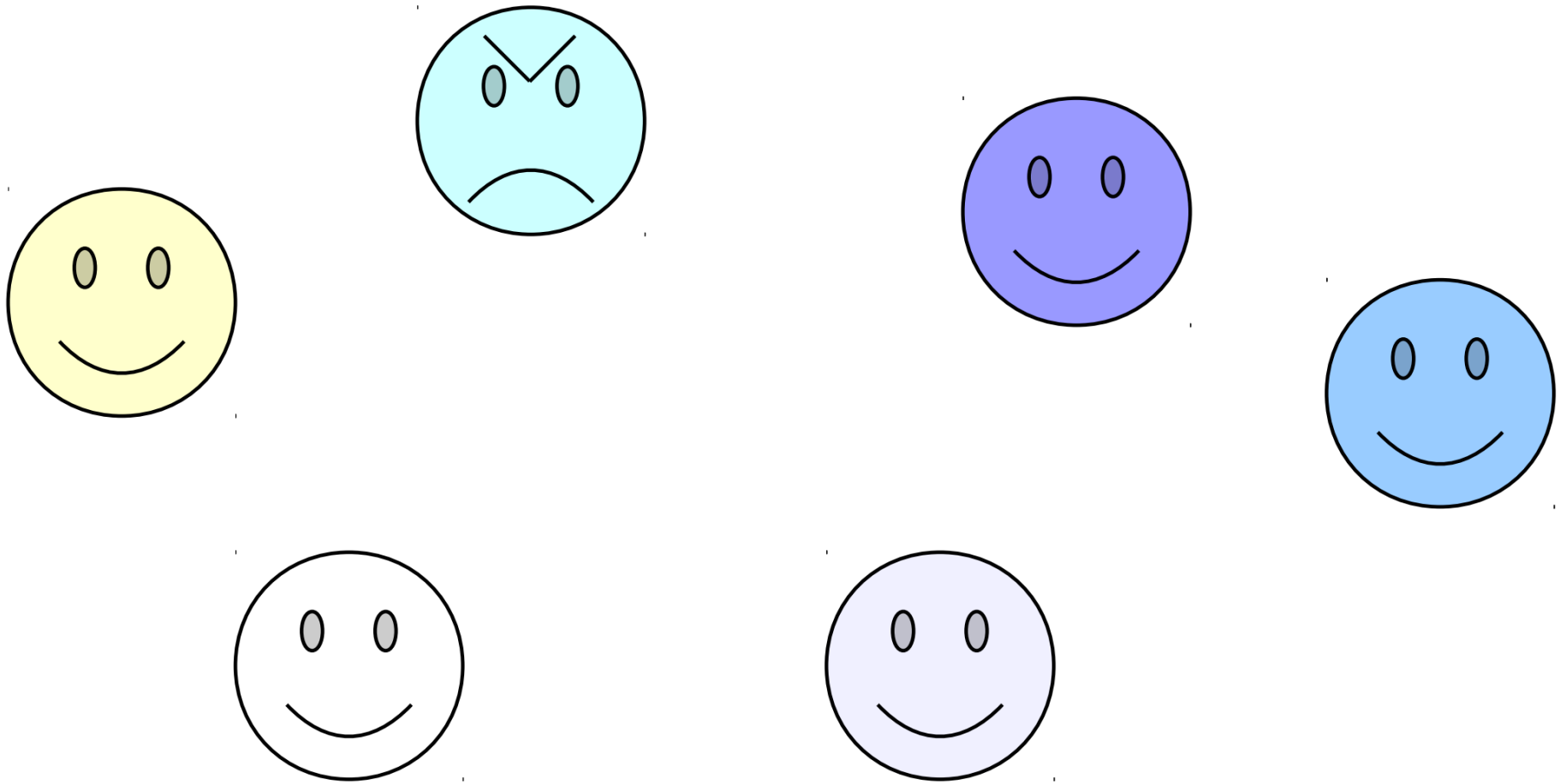


$\forall x.\ Smiling(x)$

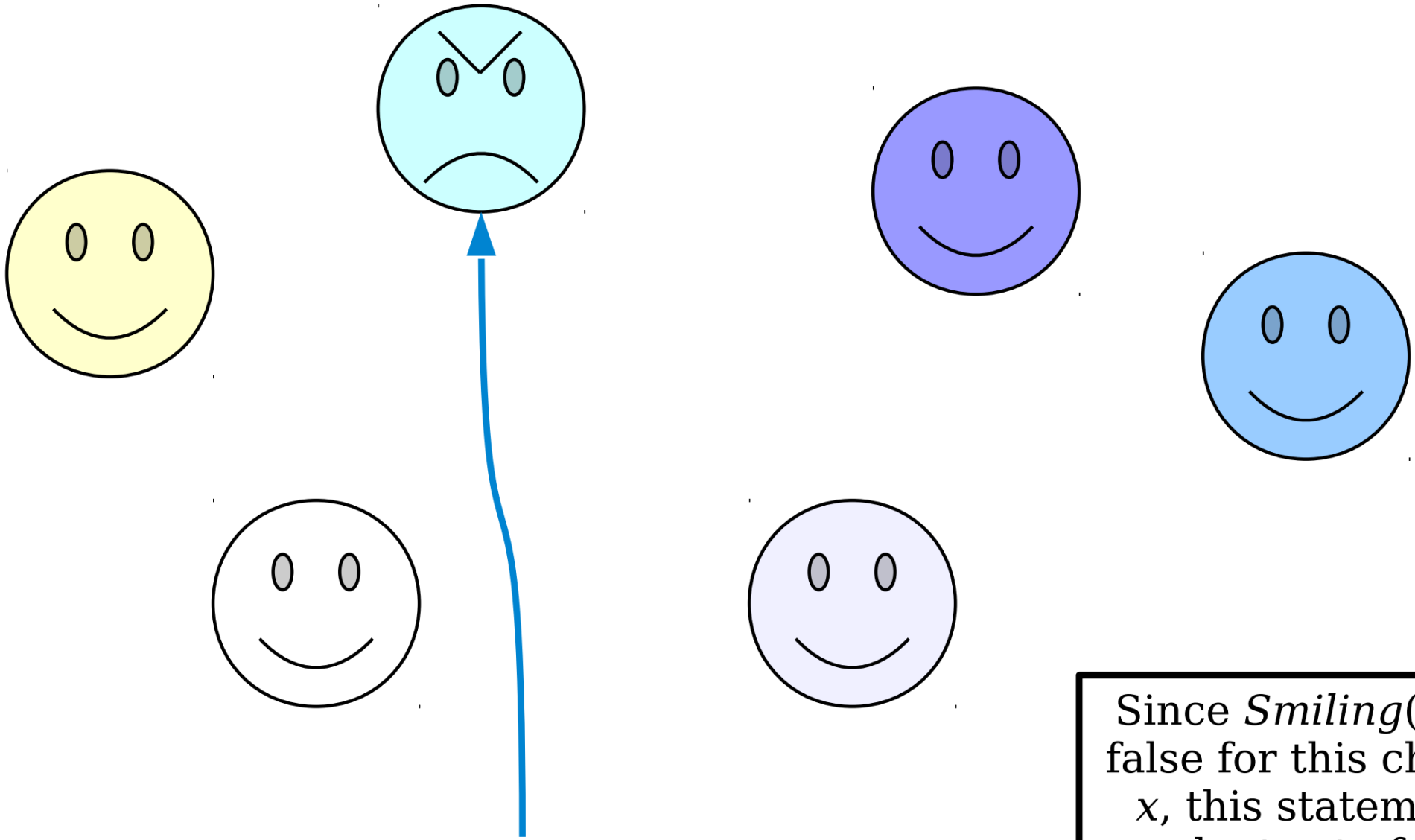


$\forall x.\ \mathit{Smiling}(x)$

Since $\mathit{Smiling}(x)$ is true for every choice of x, this statement evaluates to true.



$$\forall x.\ \mathit{Smiling}(x)$$



~~$\forall x.\ Smiling(x)$~~

Since $Smiling(x)$ is false for this choice x, this statement evaluates to false.

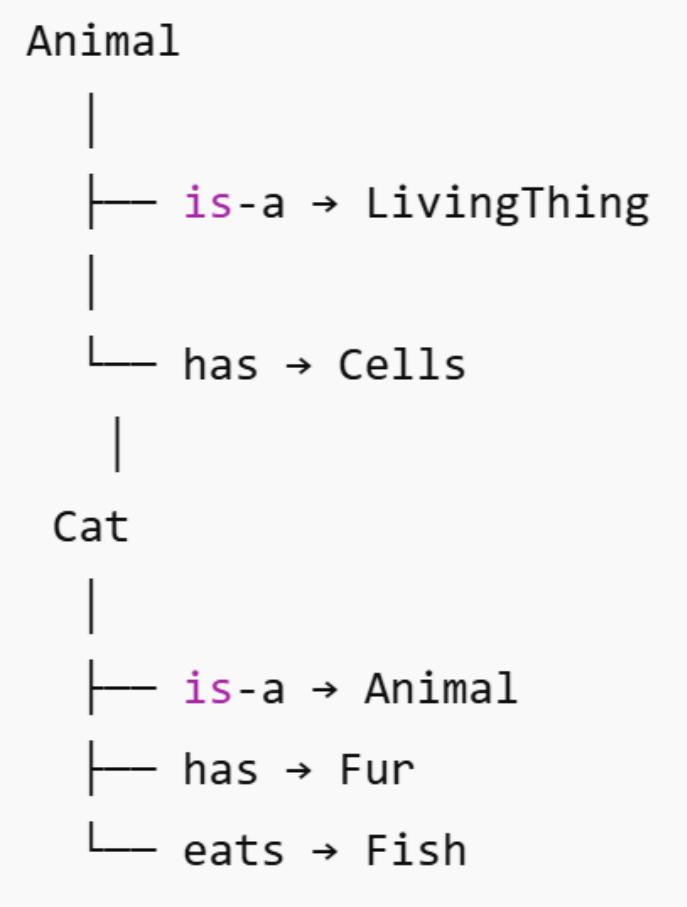
Semantic Networks

A **semantic network** is a way of representing knowledge using a **graph-like structure** where:

- **Nodes** represent *concepts, objects, or entities* (e.g., *Cat, Animal, Fur*).
- **Edges (links)** represent *relationships* between these concepts (e.g., *is-a, has, part-of*).

They are widely used in **artificial intelligence, natural language processing, and cognitive science** to model how concepts are related in human memory and reasoning.

Example



Nodes: Cat, Animal, Fur, Fish
Edges: labeled links between nodes

Applications

- **AI reasoning** (expert systems, knowledge bases).
- **NLP** (word meaning, disambiguation).
- **Ontology design** (e.g., in Semantic Web with RDF/OWL).
- **Cognitive modeling** (how humans store and retrieve knowledge).

Other logics

- Temporal logic
- Fuzzy logic

Questions?